

Challenge 1: Magnetic Levitation Control

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1 Mechanical System Model

Before establishing the control system, the mechanical system was modelled using a set of couple first-order ordinary differential equations. This was done by first modelling each electromagnet-rail system individually. Then they were considered in aggregate to model the motion of the system as a whole.

1.1 Mass Spring Models

The vibration of the rail at each electromagnet location can be modelled using four mass-spring systems where one end is stationary and the other is attached to the mass of the rail that is affected by an electromagnet (m_R). This model can be seen in Figure 1. The spring has a spring constant k_R, and the distance between the rail and electromagnet is assigned z_R.

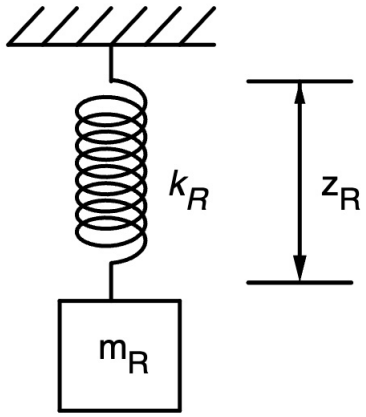


Figure 1: Mass-spring system of the rail vibration.

The electromagnet vibration can be modelled for each electromagnet using four coupled-mass spring systems. This models the deflection of the rail in addition to the motion of the electromagnet as a harmonic oscillator. The end of each spring is attached to the vehicle frame and the mass of one electromagnet (m_E) so that the system responds as the vehicle moves or changes positions. This model can be seen in Figure 2. The springs have a spring constant k_B, which represents the suspension beam for each electromagnet. The ground is defined as being z=0, the distance from the ground to the rail is z_R, and the distance from the ground to electromagnet as z_E. the gap can be defined as the difference between the position of the rail and the electromagnet, which is the sensor reading of the gap (g), as it changed in response to both moving with respect to one another. These values then yield the equations of motion.

This is the sensor reading of the gap (g) that would change in response to the movement and rotation of the vehicle.

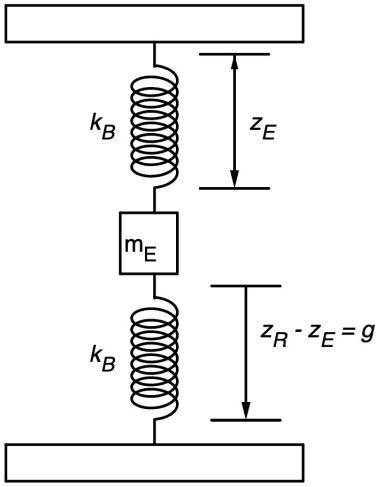


Figure 2: Mass-spring system of the electromagnet vibration.

1.2 Free Body Diagrams

The electromagnet vibration and the subsequent forces generated can be modelled using four free body diagrams for the electromagnets at the forward-left (Figure 3), forward-right (Figure 4), aft-left (Figure 5), and aft-right (Figure 6). The positive z direction is defined as pointing downwards. The force from the first spring is acting upwards and the force from the second spring is pointing downwards. The weight of the electromagnet and the attractive force of the electromagnet, which is attracted to the rail, are pointed downwards.

The vibration of the rail and their subsequent forces generated are modelled using four free body diagrams for the electromagnets at the forward-left (Figure 7), forward-right (Figure 8), aft-left (Figure 9), and aft-right (Figure 10). The positive z direction is also defined as pointing downwards. The force from the spring is acting upwards and the weight of the rail and the attractive force of the electromagnet are pointed downwards.

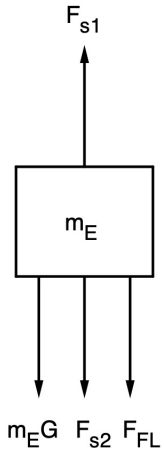


Figure 3: Free body diagram of the electromagnet vibration at the forward-left.

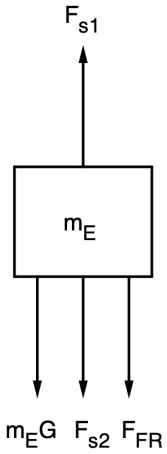


Figure 4: Free body diagram of the electromagnet vibration at the forward-right.

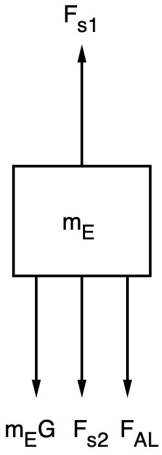


Figure 5: Free body diagram of the electromagnet vibration at the aft-left.

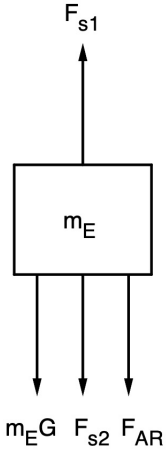


Figure 6: Free body diagram of the electromagnet vibration at the aft-right.

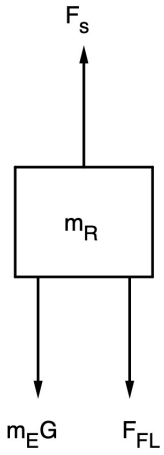


Figure 7: Free body diagram of the rail vibration at the forward-left.

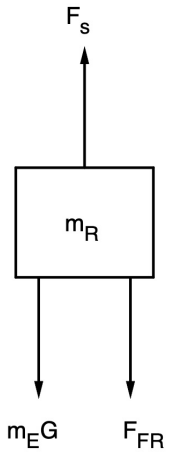


Figure 8: Free body diagram of the rail vibration at the forward-right.

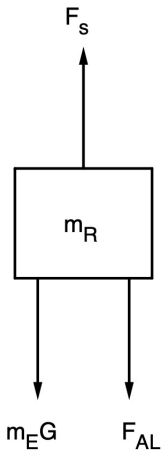


Figure 9: Free body diagram of the rail vibration at the aft-left.

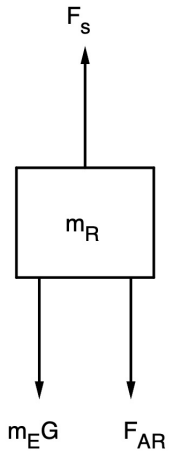


Figure 10: Free body diagram of the rail vibration at the aft-right.

1.3 Equations of Motion

The system is comprised of various equations of motion that can be derived using the previously determined free-body diagrams and mass-spring systems.

1.3.1 Rail and Electromagnet Vibration

From the free body diagram in Figure 2, Newton's second law can be used to sum all of the vertical forces generated by the vibration of the rail. ΣF_R represents the net force acting on the rail and the attractive forces for each electromagnet are defined as F_{FL}, F_{FR}, F_{AL}, F_{AR} which are in reference to their locations on the vehicle. These equations relate the vertical position of the rail and the attractive force generated by each electromagnet and can therefore be implemented in the control system.

$$\Sigma F_R = m_R \ddot{z}_R = -k_R z_R + F_{FL} + m_R G \tag{1}$$

$$\Sigma F_R = m_R \ddot{z}_R = -k_R z_R + F_{FR} + m_R G \tag{2}$$

$$\Sigma F_R = m_R \ddot{z}_R = -k_R z_R + F_{AL} + m_R G \tag{3}$$

$$\Sigma F_R = m_R \ddot{z}_R = -k_R z_R + F_{AR} + m_R G \tag{4}$$

1.3.2 Electromagnet Vibration

The free body diagram in Figure 3 and Newton's second law can also be used to sum all of the forces representing the electromagnet vibration of each of the four electromagnets. ΣF_E represents the net force from the electromagnets and the attractive forces for each electromagnet are once again defined as F_{FL}, F_{FR}, F_{AL}, F_{AR}. These equations relate both the vertical position of the electromagnet, the displacement of the electromagnet and the rail or the gap distance, and the attractive force generated by each electromagnet. These equations are also utilized in the control system.

$$\Sigma F_E = m_E \ddot{z}_E = -k_B m_E + k_B(z_R - z_E) + m_E G + F_{FL} \tag{5}$$

$$\Sigma F_E = m_E \ddot{z}_E = -k_B m_E + k_B(z_R - z_E) + m_E G + F_{FR} \tag{6}$$

$$\Sigma F_E = m_E \ddot{z}_E = -k_B m_E + k_B(z_R - z_E) + m_E G + F_{AL} \tag{7}$$

$$\Sigma F_E = m_E \ddot{z}_E = -k_B m_E + k_B(z_R - z_E) + m_E G + F_{AR} \tag{8}$$

1.3.3 Vertical Motion

To consider the motion of the vehicle as a whole, there are three parameters to account for:

- Vertical motion
- Pitch

- Roll

Obtaining the linear vertical motion of the vehicle (at the center of mass) can be done by simply summing the vertical components of the electromagnet force. This yields:

$$F_v = m_v a_v = -m_v G + F_{\epsilon,\text{net}} \tag{9}$$

where F_v is the vertical force on the vehicle, m_v is the vehicle mass, a_v is the acceleration of the vehicle vertically, and we define

$$F_{\epsilon,\text{net}} = \sum_i F_{\epsilon,i} \tag{10}$$

1.3.4 Vehicle Pitch

To model the vehicle pitch, the torque on the system must first be considered. This is done by considering the forces that have been previously described and using the definition of torque to yield:

$$\tau_x = I_x \alpha = \sum_{i=1}^{4} y_{\epsilon,i} F_{m,i} \cos\theta_i \tag{11}$$

where $y_{\epsilon,i}$ is the horizontal distance of the electromagnet from the vehicle center of mass, and θ is the angle that the connecting arm makes with the vehicle center of mass. If it can be assumed that the arm is at an angle of $\frac{\pi}{4}$ rad, then the pitch reduces to:

$$\tau_x = I_x \alpha = \frac{1}{\sqrt{2}} \sum_{i=1}^{4} y_{\epsilon,i} F_{m,i} \tag{12}$$

1.3.5 Vehicle Roll

Determining the vehicle roll is effectively the same as determining the vehicle pitch. Hence, the equation for roll is:

$$\tau_y = I_y \alpha = \sum_{i=1}^{4} x_{\epsilon,i} F_{m,i} \sin\theta_i \tag{13}$$

Using the same assumption as before, the roll equation is:

$$\tau_y = I_y \alpha = \frac{1}{\sqrt{2}} \sum_{i=1}^{4} x_{\epsilon,i} F_{m,i} \tag{14}$$

1.3.6 Parameters and Units

Table 1 summarizes all of the parameters and their corresponding units that are used in both the control and mechanical systems.

Parameter	**Units**
Gravity	[m/s^2]
Gap Distance	[m]
Forces	[N]
Mass	[kg]
Spring Constant	[N/m]
Moment of Inertia	[kg·m^2]

Table 1: The defined parameters used in all systems and their set units.

2 Control System

The control system uses PID controllers in order to regulate and control the gap distance between the electromagnet and the rail. The PID controller, which stands for Proportional Integral Derivative, uses those three parameters to correct the error seen in Equation (15) and adjust the system accordingly using the desired reference setpoint [2] [3]. Thus, the PID controller is able to stabilize the levitation by taking the input readings from the sensors and adjusting them accordingly. As there are four electromagnets that are independent of one another, each electromagnet is controlled individually by its own PID controller. The control system uses four inputs, which are the gap distances, and four outputs, which are the four forces of attraction that are generated by the electromagnets.

2.1 Equations

There are three equations that make up the control system and the PID controller. The first is Equation (15), which defines the error signal, $e_i(t)$, and the input of the PID. It is the difference between the reference setpoint, which here is the desired gap distance, and the actual gap distance that is received by the sensors located on the vehicle. Since the desired gap distance is between 5 to 10 mm, an average distance of 7.5 mm was used as the reference setpoint. This can be seen as the constant value used in the signal flow path diagram in Figure 11.

$$e_i(t) = g_{\text{ref}} - g_i \tag{15}$$

The next equation, Equation (16), calculates the gap distance that is received from the sensor readings by taking the difference between the location of the

rail, z_R, and the position of the electromagnet, z_E. This value is then used to calculate the error signal in Equation (15).

$$g_i = z_R - z_E \tag{16}$$

Finally, the last equations, Equation (17) and Equation (18), both show the output of the PID controller, or the force of attraction generated by the electromagnet. It uses three different tuning parameters following what was proposed by Ziegler and Nichols: the proportional gain (K_P), the integral gain (K_i), the derivative gain (K_d), the integral time (T_i), and the derivative time (T_d) [2]. These parameters or gains can directly impact the stability of the system and thus can be tuned in order to stabilize the system and output [2]. In the control system, as seen in the signal flow path diagram, they are contained within the PID controller block.

$$F(t) = K_P \left(e_i(t) + \frac{1}{T_i} \int e_i(t)\, dt + T_d \frac{de_i(t)}{dt} \right) \tag{17}$$

$$F(t) = K_P e_i(t) + K_i \int e_i(t)\, dt + K_d \frac{de_i(t)}{dt} \tag{18}$$

2.2 Signal Flow Path Diagram

The control system process can be modelled and seen using a closed-loop feedback control system. A closed-loop control system is when the error signal enters a controller and outputs a more desired value [2]. The flow path diagram that is seen in Figure 11 shows the closed-loop control system where the error signal enters the PID controller and the controller output enters the plant or mechanical system, which uses the readings of the sensor, to output a value (g(t)). This is then fed back into the loop and used to calculate the error signal alongside the reference setpoint. Figure 11 shows the signal flow path diagram of this process for one electromagnet or one sensor reading.

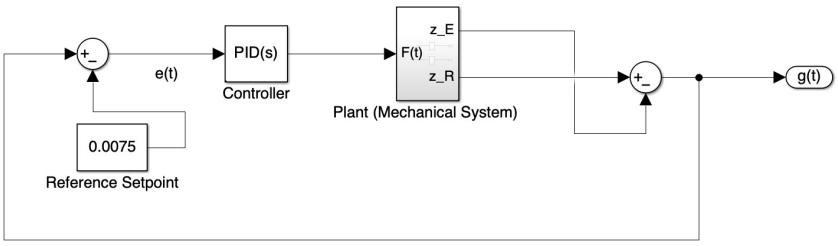


Figure 11: Closed-loop feedback control for one electromagnet.

3 Explanation and Justification

The use of magnets and electromagnets naturally results in an unstable system that can ultimately prove critical in any high-speed vehicle. This results in the need to monitor the system of the vehicle in order to ensure the stability of the vehicle and a desirable gap distance is maintained. The solution to this problem is to employ a control system that will monitor and stabalize the system. A control system was successfully developed and is subsequently summarized below.

3.1 Control System

The control system (Figure 12-15) utilizes four PID controllers, one for each electromagnet, in order to stabilize the levitation of the vehicle by altering the forces of attraction that are generated by the electromagnets. The input of the PID controllers are the error signals that are determined by both the set reference point and the actual gap of the pod. These values change independently for each electromagnet and with respect to time and thus they need to be constantly monitored and changed. The output of the PID controllers are the forces required to make such changes. These values are then fed into the mechanical systems or plants (Figure 13) that are comprised of the pre-determined equations of motion of both the rail vibration and electromagnet vibration. Thus, they take these forces and run them through the plants or mechanical systems to then calculate the necessary positions of z_E and z_R or the actual gap distance, g, that is supplied by the sensor (Figures 14-15). These values are then fed back into the feedback control system to calculate the error signal. The process continues once more as new readings are inputted by the sensors. At the same time, with the scope block, the changing gap distance is plotted for each electromagnet as it changes with time and stabilizes. As shown in Equation (17) and Equation (18), PID controllers use tuning variables which can directly

affect the stability of the system and so it is important to determine the best values for K_p, K_i, and K_d [1] [3] [2]. A beneficial aspect of simulating the control system in simulink is that the PID controller block can be set so that it automatically tunes the PID controller and chooses the most optimal gains for the output [1]. Thus, the program is capable of automatically controlling and adjusting the internal coefficients to best stabilize the system.

This process of tuning and stabilizing the system can be seen in Figure 16 where the gap for each electromagnet is plotted with respect to time. The plot shows how the gap oscillates between a small gap and then increases to the preferred gap distance of between 5-10mm when the system is stabilized. The simulation merely shows how this process would continuously repeat over and over in order to ensure the system is stable as the sensor values would be fed into the control system and plotted as the gap evolves and changes over time.

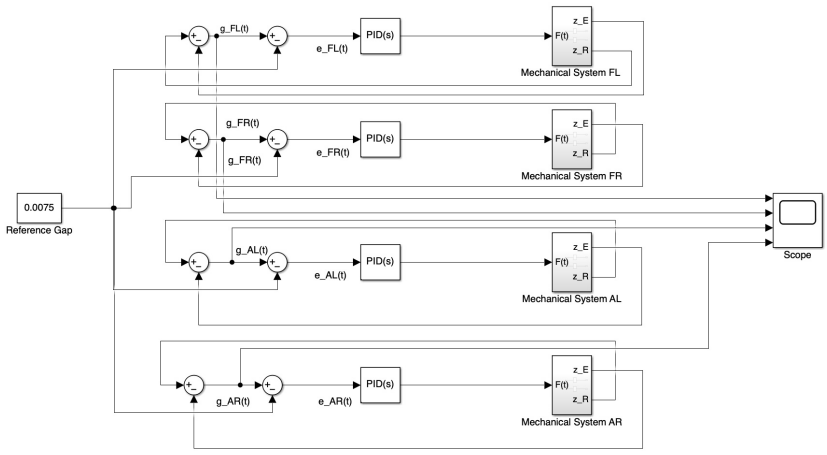


Figure 12: Control system in simulink.

3.2 Plant

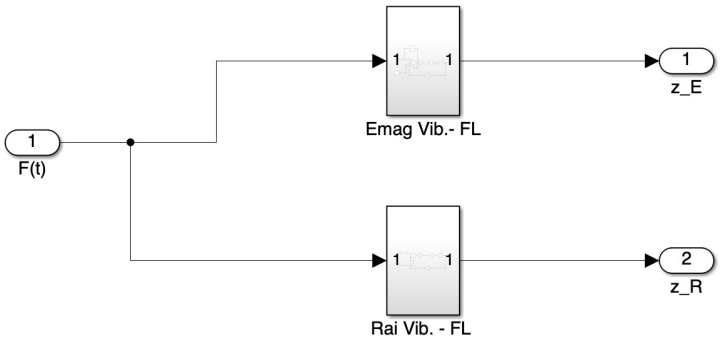


Figure 13: Plant in the control system that represents the mechanical system or equations of motion. They compute the values of z_E and z_R to find the actual gap distance. They used to calculate the actual gap distance that would be provided from the sensor readings.

3.3 Electromagnet Vibration

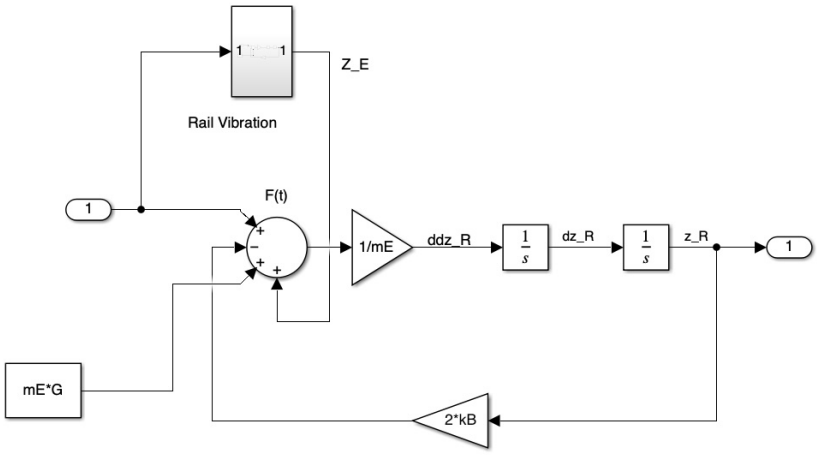


Figure 14: Model of the electromagnet vibration in the control system.

3.4 Rail Vibration

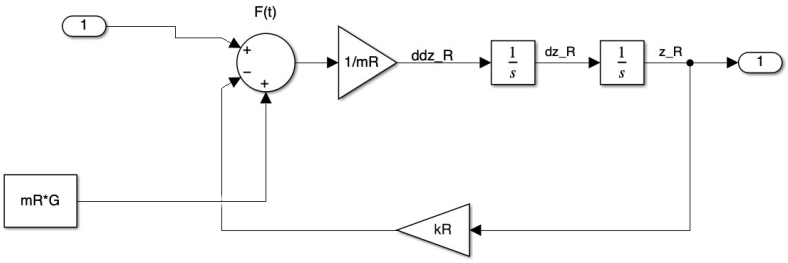


Figure 15: Model of the rail vibration in the control system. It is also used in the electromagnet vibration model.

3.5 Plot

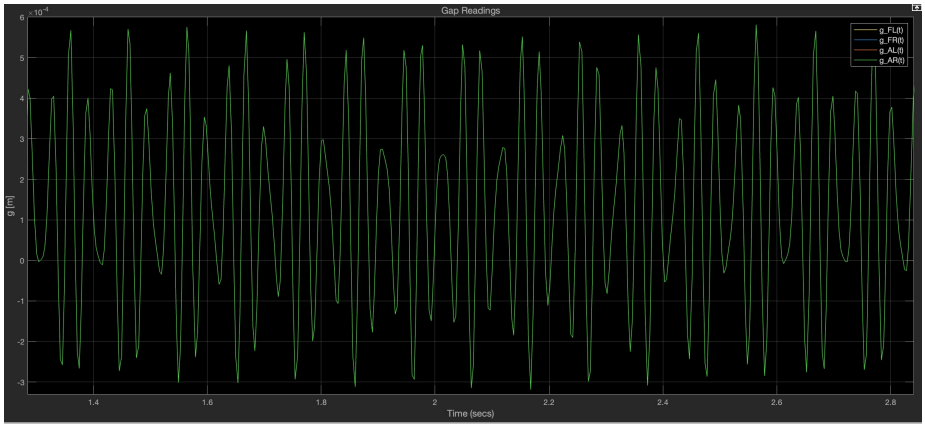


Figure 16: Plot of the output of the control system and the stabalized gap for each electromagnet with respect to time.

References

[1] *Control Tutorials for MATLAB and Simulink - Introduction: PID Controller Design*. URL: `https://ctms.engin.umich.edu/CTMS/index.php?example=Introduction&section=ControlPID` (visited on 05/11/2025).

[2] Katsuhiko Ogata. *Modern Control Engineering*. en. 5th ed. Boston: Prentice-Hall, 2010. ISBN: 978-0-13-615673-4.

[3] J Mas Soldevilla. "Dynamic Analysis and Stability Study of the Electromagnetic Suspension Levitation System of the Hyperloop". en. In: ().